Measurement-Efficient Port-Reading Policies for Fluid Antenna Systems Under Spatially Correlated κ-μ Shadowed Fading

Matheus F. Silva, Pedro M. R. Pereira, Felipe A. P. de Figueiredo

***Abstract*— Fluid antenna systems can exploit spatial diversity by selecting among many candidate ports, but practical receivers cannot always read all ports before selection. This work studies the port-reading stage itself, namely, the choice of which ports should be measured under a strict observation budget. Under spatially correlated κ-μ shadowed fading, geometry-driven uniform reading, threshold-aware marginal ranking, and greedy coverage are compared using observed-only outage as the evaluation metric. For a 100-port aperture and budgets from 3 to 7 measured ports, the results reveal a clear budget-dependent behavior. Greedy coverage is most effective when only three ports are read, winning 12 of 14 fading regimes. However, for four or more measured ports, uniform subsampling becomes the most reliable strategy and achieves the best average outage across regimes. Marginal ranking is consistently inferior in the evaluated regimes, indicating that per-port success frequency can be a poor proxy for set-level diversity in correlated apertures. These results establish port reading as a critical design layer for measurement-efficient fluid antenna operation.**

***Keywords*— Fluid antenna systems, port selection, port reading, outage probability, spatial correlation, κ-μ shadowed fading, Monte Carlo.**

I. Introduction

Fluid antenna systems (FASs) have emerged as a promising antenna technology for beyond-5G and 6G systems, particularly when size and hardware constraints limit the use of conventional multi-antenna arrays [1]. By electronically moving a single radiating element across multiple candidate positions, a FAS can exploit spatial diversity with only one radio frequency (RF) chain, which makes it attractive for compact terminals and sensor-class devices [2, 3]. When the best port is selected, the resulting diversity gain can approach or even exceed that of classical combining schemes despite the small aperture size [4]. However, these gains usually rely on reading many ports, which increases measurement overhead, switching delay, and receiver complexity.

Partial-observation FAS operation reduces this burden by measuring only a subset of ports before the final selection decision. Existing studies have investigated machine learning (ML)-based port selection, deep-learning-based transceiver strategies, and generative methods that infer unobserved ports from sparse measurements [5-9]. These approaches mainly focus on the selector or predictor used after the measurements are acquired. Nevertheless, when only a few ports can be read, the sensing pattern itself may dominate the final outage performance.

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Most prior partial-observation FAS studies, therefore, leave an important design layer underemphasized. In particular, spatial correlation makes port reading nontrivial because ports that are individually promising may provide redundant information when read together. This motivates a separate design-stage evaluation of the port-reading problem before considering any reconstruction or learning-based selector. Such an evaluation clarifies whether the measured set should prioritize geometric coverage, marginal port quality, or complementarity among ports.

This work treats port reading as a policy-level design problem for measurement-limited FAS operation. The proposed evaluation isolates the measured port set from any downstream reconstruction model through an observed-only outage metric, in which the receiver selects only among the measured ports. Under spatially correlated κ-μ shadowed fading, three low-complexity policies are compared: uniform geometric reading, threshold-aware marginal ranking, and greedy coverage. The results reveal a budget-dependent crossover, where greedy coverage is preferable for $N = 3$, whereas uniform reading becomes the most robust strategy for $N \geq 4$. Conversely, marginal ranking is consistently inferior, showing that isolated per-port quality is not a reliable surrogate for set-level diversity in correlated apertures.

The main contributions of this work are summarized as follows:

1) Port reading is formulated as a policy-level design problem for measurement-limited FAS operation, separating the measured-port choice from downstream reconstruction or selection.
2) An observed-only outage metric is adopted to isolate the effect of the measured port set under identical observation budgets and fading conditions.
3) Three low-complexity policies are compared under spatially correlated κ-μ shadowed fading: uniform reading, marginal ranking, and greedy coverage.
4) A Monte Carlo evaluation with $M = 100$ ports, 14 fading regimes, and $N \in \{3, 4, 5, 6, 7\}$ reveals that greedy coverage is best for $N = 3$, while uniform reading is more robust for $N \geq 4$.
5) The results show that per-port exceedance frequency is a weak proxy for diversity, highlighting the need to design reading policies at the measured-set level.

II. Dataset and System Model

A. Signal Model and Partial Observation

A downlink scenario is considered, as illustrated in Fig. 1, in which a single-antenna base station (BS) serves a terminal

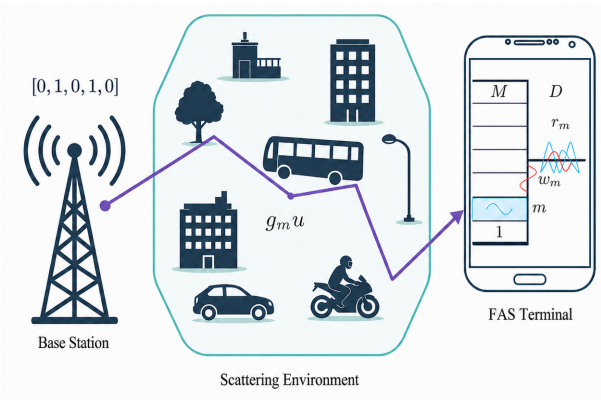

Fig. 1. Downlink FAS model.

equipped with an FAS composed of M candidate ports distributed over an aperture of electrical length D. Let $u \in \mathbb{C}$ denote the transmitted symbol with average power $\mathbb{E}[|u|^2] = P_u$. The baseband sample collected at port $m \in \{1, \ldots, M\}$ is modeled as

$$r_m = g_m u + w_m, \tag{1}$$

where g_m is the complex channel coefficient associated with port m, and $w_m \sim \mathcal{CN}(0, N_0)$ is circularly symmetric complex Gaussian noise. The instantaneous signal-to-noise ratio (SNR) at port m is therefore

$$\eta_m = \frac{P_u}{N_0}|g_m|^2, \tag{2}$$

and the full SNR profile is collected in $\boldsymbol{\eta} = [\eta_1, \ldots, \eta_M]^\mathsf{T}$.

To reduce measurement overhead, only a subset of ports is observed. Let $\mathcal{O}$ denote the index set of measured ports with $|\mathcal{O}| = N \ll M$, and define the observation vector $\boldsymbol{\eta}_\mathcal{O} = [\eta_{m_1}, \ldots, \eta_{m_N}]^\mathsf{T}$ for $\mathcal{O} = \{m_1, \ldots, m_N\}$. Under partial observation, any observed-only selector or profile-reconstruction method must operate from $\boldsymbol{\eta}_\mathcal{O}$ rather than from the full profile $\boldsymbol{\eta}$.

B. Spatially Correlated κ-μ Shadowed Fading

The ports are assumed to experience spatial correlation due to sub-wavelength sampling along the aperture. For isotropic 2D scattering in the azimuthal plane, the correlation between ports m and n is modeled through a Jakes-type matrix $\mathbf{R} \in \mathbb{C}^{M \times M}$ with entries

$$[\mathbf{R}]_{m,n} = J_0\left(\frac{2\pi|m-n|}{D(M-1)}\right), \tag{3}$$

where $J_0(\cdot)$ denotes the zeroth-order Bessel function of the first kind. Following [9], let the eigendecomposition be $\mathbf{R} = \mathbf{V}\boldsymbol{\Sigma}\mathbf{V}^\mathsf{H}$, and define the correlation-shaping matrix $\mathbf{B} = \mathbf{V}\boldsymbol{\Sigma}^{1/2}$.

To capture both multipath clustering and shadowing on dominant components, the port-wise fading is constructed by adapting the physical κ-μ shadowed model to a spatially correlated multiport vector process. In particular, the noncoherent cluster superposition in (4) follows the canonical model definition and is rewritten here with an explicit port index and correlation shaping [10, eq. (7)]. A physically motivated construction expresses the received power at port m as a sum over μ clusters,

$$|g_m|^2 = \sum_{\ell=1}^{\mu} |q_{m,\ell} + p_m|^2, \tag{4}$$

where $q_{m,\ell}$ represents the scattered component of cluster ℓ at port m, and p_m is the dominant component shared across clusters at that port. Spatially correlated scattered vectors are generated as $\mathbf{q}_\ell = \sigma_q \mathbf{B}\mathbf{z}_\ell$, where $\mathbf{q}_\ell = [q_{1,\ell}, \ldots, q_{M,\ell}]^\mathsf{T}$, $\mathbf{z}_\ell \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_M)$, and σ_q^2 controls the average scattered power.

The dominant component vector is modeled as $\mathbf{p} = \xi\,\boldsymbol{\upsilon}$, where $\boldsymbol{\upsilon} \in \mathbb{C}^M$ is a deterministic steering vector with identical entries υ_0, and ξ is a Nakagami-m random variable with $\mathbb{E}[\xi^2] = 1$ that accounts for shadowing. Since all ports are identically distributed, the ratio of dominant to scattered power is enforced by choosing $|\upsilon_0|^2 = \kappa\sigma_q^2$, equivalently $\kappa = |\upsilon_0|^2/\sigma_q^2$. The parameters κ, μ, and m retain their standard interpretations as the dominant-to-scattered power ratio, the number of clusters, and the shadowing severity, respectively [11].

C. Port-Selection Metric

With full observation, the ideal FAS selects the port that maximizes the instantaneous SNR,

$$\eta^\star = \max_{1 \le m \le M} \eta_m. \tag{5}$$

Given a threshold η_{th}, the outage probability (OP) is defined as

$$P_{\text{out}} = \Pr(\eta^\star < \eta_{\text{th}}), \tag{6}$$

and is approximated through Monte Carlo evaluation over independent channel realizations.

Under partial observation, the observed-only selector chooses the best port among the measured indices,

$$\eta^{\text{obs}} = \max_{m \in \mathcal{O}} \eta_m, \tag{7}$$

which yields $P_{\text{out}}^{\text{obs}} = \Pr(\eta^{\text{obs}} < \eta_{\text{th}})$. This metric intentionally isolates the impact of the port-reading policy from the effects of any particular reconstruction model. Alternatively, a reconstruction-aided selector may first estimate the full profile $\widehat{\boldsymbol{\eta}}$ from $\boldsymbol{\eta}_\mathcal{O}$, and then select $\widehat{m} = \arg\max_m \widehat{\eta}_m$. This second stage is not optimized in this work, since the objective is to evaluate the measured port set itself.

D. Dataset Construction

A multi-fading dataset is generated to promote robustness across heterogeneous conditions. For each parameter set (κ, μ, m) in a grid of 14 configurations spanning Rayleigh, Nakagami, Rician, κ-μ, and its shadowed distributions, 10^6 independent realizations of $\boldsymbol{\eta}$ are produced using $M = 100$ ports and electrical length $D = 1$ [12]. For each realization, the observed index set $\mathcal{O}$ is generated according to a port-reading policy and an observation budget $N \in \{3, 4, 5, 6, 7\}$. The complete set of parameter combinations is summarized in Table I[1].

[1]Due to numerical stability, the values 0.001 and 50 denote limiting cases tending to 0 and ∞, respectively.

TABLE I
FADING MODEL PARAMETERS USED IN THE DATASET.

Dataset	κ	μ	m	Channel distribution
1	0.001	1	0.001	Rayleigh, severe shadowing
2	0.001	1	2	Rayleigh, moderate shadowing
3	0.001	1	50	Rayleigh
4	0.001	2	50	Nakagami
5	0.001	5	50	Nakagami
6	5	1	0.001	Rician, severe shadowing
7	5	1	2	Rician, moderate shadowing
8	5	1	50	Rician
9	5	2	0.001	κ-μ, severe shadowing
10	5	2	2	κ-μ, moderate shadowing
11	5	2	50	κ-μ
12	5	5	0.001	κ-μ, severe shadowing
13	5	5	2	κ-μ, moderate shadowing
14	5	5	50	κ-μ

III. PORT-READING POLICIES AND EVALUATION FRAMEWORK

A. Reconstruction Context and Observed-Only Evaluation

Under partial observation, some transceiver strategies attempt to reconstruct the full vector $\boldsymbol{\eta} \in \mathbb{R}^M$ from the measured entries $\boldsymbol{\eta}_{\mathcal{O}} \in \mathbb{R}^N$ before final port selection. This stage can be expressed through a mapping

$$\widehat{\boldsymbol{\eta}} = f\left(\boldsymbol{\eta}_{\mathcal{O}}\right), \tag{8}$$

where f may represent any estimator, including parametric ML regressors or model-based interpolators. When supervised learning is adopted, f is commonly fit by minimizing a reconstruction loss over a training set, for instance, the mean squared error

$$\mathcal{L}_{\mathrm{MSE}} = \frac{1}{M} \sum_{m=1}^{M} \left(\widehat{\eta}_m - \eta_m\right)^2. \tag{9}$$

Although the choice of f governs the achievable reconstruction fidelity, the port-reading pattern $\mathcal{O}$ defines which information is available at the input and therefore constrains the downstream decision. This work intentionally evaluates observed-only outage to isolate the reading policy from the effects of any particular reconstruction model. Consequently, the reported results should be interpreted as a controlled design-stage comparison of measurement patterns rather than as an optimization of a complete reconstruction-aided receiver.

B. Port-Reading Strategies

Port reading is defined as the selection of a subset of ports to probe under a fixed observation budget. The study compares three low-complexity port-reading strategies under the same observation budgets. Policies B and C naturally provide an ordered list of ports. Policy A provides a deterministic geometry-driven subset for each budget.

Policy A is a geometry-driven baseline that uniformly subsamples the aperture. For a budget N, the selected indices are evenly spaced between ports 1 and M. This strategy prioritizes spatial coverage and requires no data-driven tuning.

Policy B is a threshold-aware marginal ranking policy learned from a design subset. Let $\mathcal{D}_{\mathrm{des}}$ denote the set of design samples and define the exceedance indicator $b_{i,m} = \mathbf{1}\{\eta_m^{(i)} > \eta_{\mathrm{th}}\}$. Policy B scores each port by its empirical exceedance frequency,

$$\widehat{p}_m = \frac{1}{|\mathcal{D}_{\mathrm{des}}|} \sum_{i \in \mathcal{D}_{\mathrm{des}}} b_{i,m}, \tag{10}$$

and ranks ports in decreasing order of $\widehat{p}_m$. Deterministic ties are resolved by the design-set mean SNR at each port, and then by the port index. This policy favors ports that are frequently above threshold when considered individually.

Policy C is a greedy coverage policy that accounts for complementarity at the measured-set level. Starting from an empty set, the policy selects one port per step to maximally reduce observed-only outage on the design split. At step k, let $\mathcal{S}_{k-1}$ denote the already selected ports, and let $\mathcal{U}_{k-1}$ denote the design samples that remain uncovered, meaning that all currently selected ports fail the threshold. The next port is selected as

$$m^\star = \arg\max_{m \notin \mathcal{S}_{k-1}} \sum_{i \in \mathcal{U}_{k-1}} b_{i,m}, \tag{11}$$

which maximizes the number of newly covered samples. If all design samples become covered before the target order is completed, the remaining positions are filled using the Policy B ranking among the unselected ports. This rule keeps the output deterministic while retaining the same threshold-aware fallback criterion.

C. Per-Regime Monte Carlo Evaluation Protocol

The port-reading policies are evaluated independently within each of the 14 fading scenarios using a repeated-split Monte Carlo protocol. In each repetition, the regime samples are randomly permuted with a deterministic seed and split into a design subset and an evaluation subset using an $80/20$ fraction. Policies B and C are learned on the design subset and then evaluated on the disjoint evaluation subset. Policy A is deterministic, yet it is evaluated on the same evaluation samples to enable paired comparisons.

The primary metric is the observed-only outage probability induced by the read ports, defined for a given index set $\mathcal{O}$ as

$$\widehat{P}_{\mathrm{out}}^{\mathrm{obs}}(\mathcal{O}) = \frac{1}{|\mathcal{D}_{\mathrm{eval}}|} \sum_{i \in \mathcal{D}_{\mathrm{eval}}} \mathbf{1}\left\{\max_{m \in \mathcal{O}} \eta_m^{(i)} < \eta_{\mathrm{th}}\right\}. \tag{12}$$

For reference, the ideal baseline uses all M ports in the maximization. The threshold $\eta_{\mathrm{th}} = 0.8$ is used as a fixed operating point to compare the relative behavior of the reading policies under identical outage conditions. Since (12) depends only on exceedance events, the evaluation can equivalently operate on a binary threshold matrix, which preserves the metric exactly while simplifying the greedy computations in Policy C.

Uncertainty is characterized through the repetition-to-repetition variability induced by the repeated design/evaluation splits. Results are summarized by per-budget mean outage values across the R repetitions within each regime. To compare Policies A and C more directly, the paired difference

$$\Delta_r(N) = \widehat{P}_{\mathrm{out}}^{\mathrm{obs}}(\mathrm{C}; r, N) - \widehat{P}_{\mathrm{out}}^{\mathrm{obs}}(\mathrm{A}; r, N) \tag{13}$$

Algorithm 1: MONTE CARLO EVALUATION OF PORT-READING POLICIES

Require: Regime dataset $\mathcal{D} = \{\boldsymbol{\eta}^{(i)}\}_{i=1}^{S}$ with $\boldsymbol{\eta}^{(i)} \in \mathbb{R}^M$; threshold η_{th}; budgets $\mathcal{N}$; repetitions R; evaluation fraction ρ.

for *each regime* **do**
  for $r \leftarrow 1$ **to** R **do**
    randomly split $\mathcal{D}$ into design $\mathcal{D}_{\text{des}}$ and evaluation $\mathcal{D}_{\text{eval}}$ with fraction ρ
    compute exceedance indicators $b_{i,m} = \mathbf{1}\{\eta_m^{(i)} > \eta_{\text{th}}\}$ on $\mathcal{D}_{\text{des}}$
    learn Policy B ranking by sorting ports according to $\widehat{p}_m$ with deterministic ties
    learn Policy C order via greedy coverage on $\mathcal{D}_{\text{des}}$, with Policy B tie-breaks
    for *each* $N \in \mathcal{N}$ **do**
      construct the Policy A subset for budget N
      evaluate $\widehat{P}_{\text{out}}^{\text{obs}}(\mathcal{O}(N))$ on $\mathcal{D}_{\text{eval}}$ for Policies A–C
      evaluate the ideal outage on $\mathcal{D}_{\text{eval}}$ using all M ports
  aggregate repetition outputs into per-budget means and variability summaries across repetitions

TABLE II
PERCENTAGE OF REGIMES WON BY EACH POLICY AT EACH BUDGET.

Policy	$N = 3$ (%)	$N = 4$ (%)	$N = 5$ (%)	$N = 6$ (%)	$N = 7$ (%)
A	14.3%	85.7%	92.9%	100.0%	92.9%
B	0.0%	0.0%	0.0%	0.0%	0.0%
C	85.7%	14.3%	7.1%	0.0%	7.1%

is computed within each repetition r on the same evaluation subset. The full evaluation procedure is summarized in Algorithm 1.

IV. RESULTS AND DISCUSSION

The Monte Carlo comparison follows the protocol described in Section III-C. Each of the 14 fading regimes is processed independently with $R = 100$ repetitions, an $80/20$ design/evaluation split, and budgets $N \in \{3, 4, 5, 6, 7\}$. The outage threshold is fixed to $\eta_{\text{th}} = 0.8$ in the linear domain. For each repetition and budget, the observed-only outage of Policies A–C is computed on the evaluation samples, together with the ideal baseline that assumes all $M = 100$ ports are observable.

The regime-wise winners are summarized in Table II and visualized in Fig. 2. Policy C is strongest at the tightest budget and wins in 12 out of 14 regimes at $N = 3$. For $N \geq 4$, Policy A becomes the dominant strategy and is the unique winner at $N = 6$. Policy B never attains the lowest mean outage in the evaluated regimes.

Fig. 3 reports the complementary view in terms of the percentage of evaluation samples where the observed set contains at least one port above threshold, $100 \times (1 - \widehat{P}_{\text{out}}^{\text{obs}})$. This success-rate representation confirms the same trend. Policy A provides the highest mean above-threshold rate for $N \geq 4$, while Policy C remains strongest at $N = 3$.

Fig. 4 highlights the budget-dependent crossover between Policies A and C through the per-dataset paired mean difference in observed-only outage for $N = 3$ and $N = 4$. At $N = 3$, the differences are predominantly negative, which favors Policy C in most datasets. At $N = 4$, the sign becomes predominantly positive, which indicates that Policy A becomes preferable in most cases once one additional port can be read.

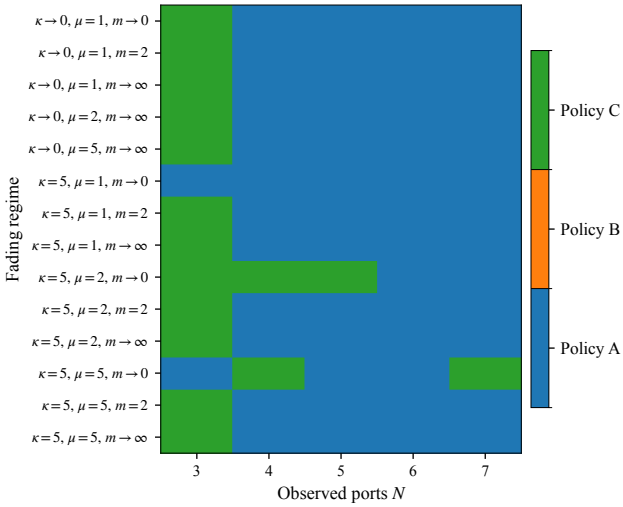

Fig. 2. Best port-reading policy by fading regime and observation budget.

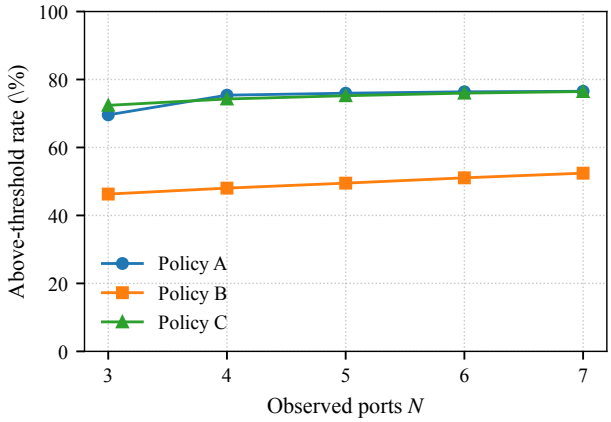

Fig. 3. Mean percentage of evaluation samples above threshold across all regimes.

Fig. 5 illustrates the typical behavior in a representative regime. As expected, all policies improve with increasing N, because additional reads provide more opportunities to observe a port above threshold. Nonetheless, Policies A and C approach the ideal baseline much faster than Policy B. The poor performance of marginal ranking indicates that, in strongly correlated and statistically homogeneous apertures, empirical per-port exceedance frequency alone is not a reliable criterion. The relevant design variable is not the isolated quality of a port, but the complementarity of the measured set.

To assess the stability of the reported trends, 95% intervals were computed from the repeated-split outputs. For regime-specific curves, the intervals are computed across the $R = 100$ design/evaluation splits and therefore quantify split-to-split variability under a fixed fading regime. For the aggregate above-threshold-rate curve in Fig. 3, repetitions are first averaged within each fading regime, and the remaining variability is computed across the 14 regimes to reflect channel-condition heterogeneity. In the representative regime of Fig. 5, this split-to-split variability is much smaller than the differences between policies. At $N = 3$, Policy C achieves 0.0239 ± 0.0001 observed-only outage, whereas Policy A obtains 0.0814 ± 0.0001. At $N = 4$, the ordering reverses, with Policy A reaching 0.0068, with interval half-width below 10^{-4}, and Policy C reaching 0.0135 ± 0.0001. Thus, the

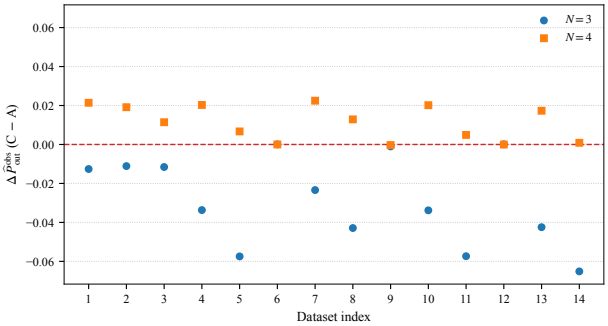

Fig. 4. Forest plot of per-dataset paired mean differences in observed-only outage between Policies C and A for $N = 3$ and $N = 4$. Negative values favor Policy C.

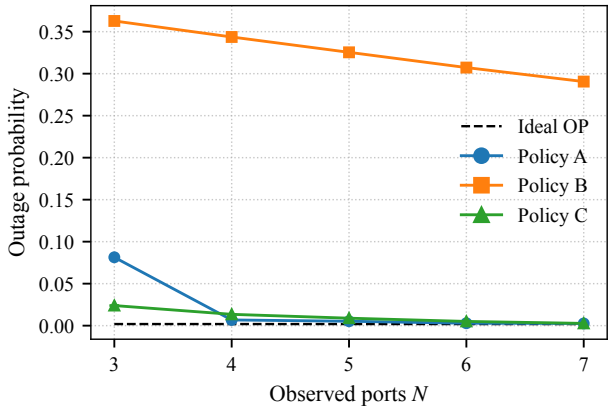

Fig. 5. Observed-only outage versus number of observed ports in the $\kappa \to 0$, $\mu = 5$, $m \to \infty$ regime.

observed crossover is not explained by random split variability.

Overall, the results show that port reading should be treated as a design layer rather than as a secondary implementation detail. Greedy coverage is useful in the extremely measurement-limited regime because it explicitly seeks complementary ports. However, once a modest number of ports can be observed, uniform spatial sampling becomes a robust and simple default. This budget-dependent crossover is the main practical insight of the study.

V. Conclusion

This work studied port reading as a distinct design layer in measurement-limited fluid antenna system (FAS) operation. A controlled observed-only outage benchmark was used to isolate the measured port set from any downstream reconstruction model. Under spatially correlated κ-μ shadowed fading with $M = 100$ ports, the repeated-split Monte Carlo evaluation across 14 regimes showed that the best port-reading strategy depends strongly on the measurement budget.

The main design message is that complementarity-aware greedy coverage provides the best protection against outage when the receiver can read only three ports. Once four or more ports are available, simple uniform reading becomes the most robust default across the evaluated fading regimes. Conversely, threshold-aware marginal ranking is consistently weak, which shows that per-port exceedance frequency is not a reliable proxy for set-level diversity in correlated apertures. Therefore, in measurement-efficient FAS design, port reading should not be treated as a secondary implementation detail, but as a core component of the selection architecture.

The results are specific to the considered aperture, threshold, and spatial-correlation model, but the evaluation framework can be directly reused for other operating points. Future work can extend this framework to reconstruction-aided selection, adaptive online port reading, and joint optimization of sensing policies under practical switching, latency, and hardware constraints.

Acknowledgments

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